INVERTIBLE MODULES ON A FINITE MODIFICATION OF A DENSE OPEN OF A NOETHERIAN DIMENSION-1 SCHEME COME FROM THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing sentence is 'Some details omitted.' and concerns only the final assembly described in the last obligation; both cited lemmas are delivered with their own proofs.

Allowed standard prerequisites (editorial).

- Surjection on Pic for finite birational modifications of dimension-1 schemes. delivered companion lemma-surjective-pic-birational-finite. First leg on X -> Z.
- Lifting invertible modules from a dimension-1 Noetherian base along a finite modification of its reduction. delivered companion lemma-lift-line-bundle-from-reduction-dimension-1. Second leg on Z -> Y.

Source reference. Original lemma 0C0T: sources/stacks-chow/tags.txt, source lines 15098–15098. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09NY: sources/stacks-chow/tags.txt, source lines 12231–12231. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume Y is Noetherian of dimension ≤ 1, f is finite, and there exists a dense open $V \subset Y$ such that $f^{-1}(V) \to V$ is a closed immersion. Then every invertible $\mathcal{O}_X$-module is the pullback of an invertible $\mathcal{O}_Y$-module.*

Proof. We factor f as $X \to Z \to Y$ where Z is the scheme theoretic image of f. Then $X \to Z$ is an isomorphism over $V \cap Z$ and Lemma 0C0T applies. On the other hand, Lemma 09NY applies to $Z \to Y$. Some details omitted. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.